MILITARY SPECIFICATION SHEET

SCANNER, MECHANICAL MX-9872(V)4/UA

This specification sheet is approved for use by USAERADCOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense. The complete requirements for procuring the MX-9872(V)4/UA Scanner shall consist of this document and MIL-S-49166.

1. SCOPE

1.1 Scope. This specification sheet covers the Mechanical Scanner, MX-9872(V)4/UA and augments MIL-S-49166. The MX-9872(V)4/UA is used on the Chaparral FLIR Subsystem, Part Number 13153184.

1.2 Purpose. The purpose of this specification sheet is to modify or expand the contents of MIL-S-49166 to describe the Scanner Assembly. All the MIL-S-49166 contents apply except as specifically modified herein. Paragraph numbers directly correlate between MIL-S-49166 and this specification sheet.

2. APPLICABLE DOCUMENTS

2.1 Order of precedence. In the event of a conflict between the text of this specification sheet and the references cited herein, the text of this specification sheet shall take precedence.

SPECIFICATIONS

MILITARY

MIL-S-49166 Scanner, Mechanical, MX-9872(V)/UA, General Specification for

DRAWINGS

18876	13144390	Scanner Assembly
80063	DLSM-B-773232	Scan-Interlace, PL-1408/UA

3. REQUIREMENTS

3.1 Description. The MX-9872(V)4/UA Scanner, herein known as the Scanner, shall consist of the MIL-S-49166, MX-9872(V)/UA Scanner adjusted for 60 Hz operation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, USA Electronics Research and Development Command, ATTN: DRDEL-E, 2800 Powder Mill Road, Adelphi, MD 20783, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.2 Construction. The MX-9872(V)4/UA shall be constructed in accordance with drawing 18876-13144390.

3.2.1 Weight. Weight of the Scanner shall be 1.55 pounds maximum.

3.6 Performance characteristics. The characteristics specified herein shall apply as defined in 3.1 and 3.6.1 except 4.6.8, 4.7.4, and 4.7.5 applies with the phase shift lens, counterweight and associated hardware removed. Operation shall be at 60 Hz using the Scan-Interlace, PL-1408/UA (DLSM-B-773232).

3.6.1 Scan angle and frequency.

3.6.1.1 Variable scan frequency. Not applicable.

3.6.1.2 Specific scan frequency (60 Hz). Scan frequency of the Scanner shall be adjusted to 60.0, ±0.5 Hz. Scan frequency drift shall not be greater than ±6.0 Hz from the frequency at ambient setting (see MIL-S-49166, 4.4) when exposed to high and low temperature environmental conditions (see MIL-S-49166).

3.6.1.3 Synchronous scan frequency. The scanner shall synchronize with a stable external synchronization source, such as the TV camera vertical sync pulse. Scan frequency shall be stable within 1 Hz of the input sync pulse frequency.

3.6.1.4 Scan angle. The scan angle shall be 12.2, ±0.5 degrees.

3.6.2 Interlace. The scanner shall operate in a 2:1 interlace mode. The interlace switching time shown in figure 1 of MIL-S-49166 shall not be greater than 2.5 milliseconds (ms).

3.6.3 Scan direction. The Scanner shall show the waveform shapes and phase relationships in figure 1 while operaing in the 2:1 rear interlace mode.

3.6.4 Power consumption. Power required to operate the Scanner shall not exceed 7.5 watts.

3.6.5 Scan efficiency. The Scanner scan efficiency shall be a minimum of 70 percent.

3.6.6 Scan jitter in azimuth. Jitter time shall be 12.5 microseconds maximum.

4. QUALITY ASSURANCE PROVISIONS

4.6 Test methods. Tests shall be performed utilizing the DLSM-B-773232 Scan-Interlace with the Scanner with a scan angle of 12.2, ±0.5 degrees at the 60.0, ±0.5 Hz scan rate.

4.6.1 Scan frequency.

4.6.1.1 Variable scan frequency. Not applicable.

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4.6.1.2 Specific scan frequency (60 Hz). The Scanner shall be connected to the Scan-Interlace, DLSM-B-773232, and mounted to a suitable fixture. Power shall be applied and the output of the azimuth (scan) transducer (Scan-Interlace DLSM-B-773232) shall be monitored with a calibrated frequency counter. The output of the Scan-Interlace (DLSM-B-773232) shall be adjusted to obtain the specific scan frequency. Failure to meet the requirements of 3.6.1.2 shall constitute failure of this test.

4.6.1.3 Synchronous scan frequency. The Scanner shall be connected to Scan-Interlace DLSM-B-773232 and mounted in a suitable fixture. Synchronization from an external stable source (such as a TV camera vertical sync pulse) and power shall be applied and the output of the azimuth (scan) transducer (Scan-Interlace DLSM-B-773232) shall be monitored with a calibrated frequency counter. Failure to meet the requirements of 3.6.1.3 shall constitute failure of this test.

4.6.1.4 Scan angle. See MIL-S-49166.

4.6.2 Interlace. See MIL-S-49166.

4.6.3 Scan direction. With the Scanner operating, the outputs of the azimuth and interlace position transducers (Scan-Interlace DLSM-B-773232) shall be observed with an oscilloscope. The output waveforms shall show the shape and phase relationships shown in figure 1. Failure to meet the requirements of 3.6.3 shall constitute failure of this test.

4.6.4 Power consumption. See MIL-S-49166.

4.6.5 Scan efficiency. See MIL-S-49166.

4.6.6 Scan jitter in azimuth. See MIL-S-49166.

6. NOTES

6.1 Intended use. The Scanner is intended for use with the Chaparral FLIR Subsystem.

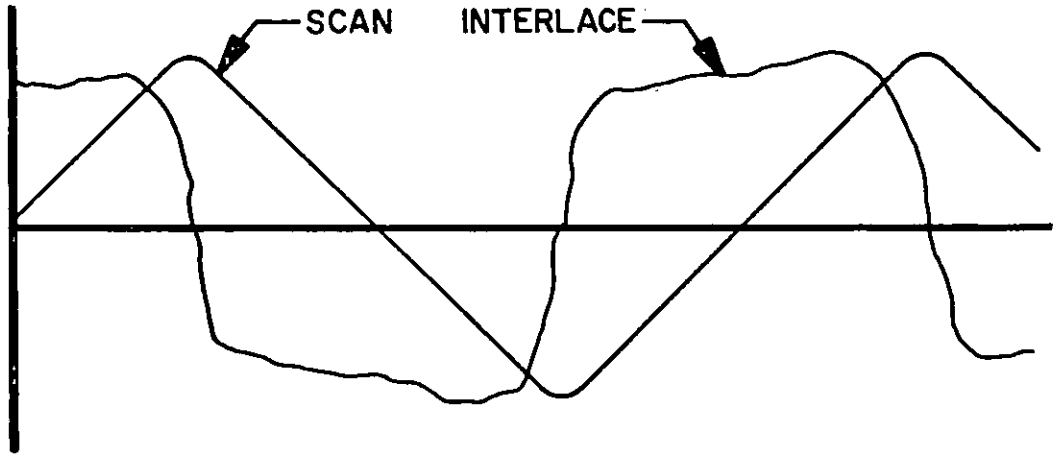
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2:1 REAR POSITION
INTERLACE LEADS SCAN APPROX. 90°



NOTES:

PL-1408/UA- CONNECT P1-33 TO P1-34 AND P1-4 TO P1-17

OSCILLOSCOPE- CONNECT CHANNEL 1 (SCAN POSITION) HIGH INPUT TO J3 (OR J2-7) AND LOW SCOPE TO GND

- CONNECT CHANNEL 2 (SCAN INTERLACE) HIGH INPUT TO J6 (OR J1-9) AND LOW SCOPE TO GND

FIGURE 1. INTERLACE TRANSDUCER AND AZIMUTH POSITION TRANSDUCER OUTPUTS.

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